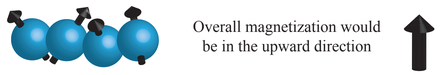
Time Dependence of the Magnetic After Effect in Co/Pd Multilayer Thin Films

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INTRODUCTION

- Hard drives record data in magnetization patterns in a thin film.
- Magnetization is related to the orientation of the electron spins within the material.



- These spins are thermally agitated and therefore contribute to the overall magnetic noise in the films magnetization.
- Current hard drives write patterns in the disk with magnetization perpendicular to the plane.



Figure 1. Above figure depicts perpendicular magnetization. Arrows point toward magnetic north.

- One way to measure the magnetization of the sample is by the Extraordinary Hall Effect (EHE).
- The EHE is a phenomenon which creates a voltage perpendicular to the current.

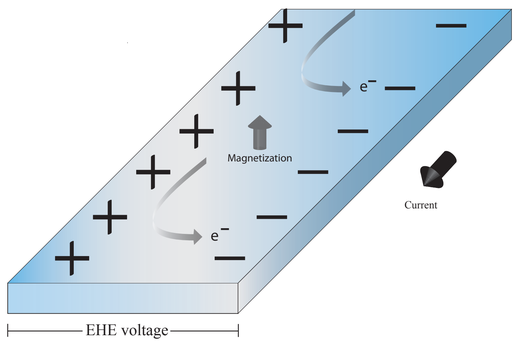


Figure 2. The above figure illustrates the relation between the EHE voltage and magnetization. As current flows perpendicular to the materials magnetization, the electrons are deflected and create a potential difference which is the EHE voltage. The stronger the magnetization, the larger the EHE voltage.

- Flipping the sample's magnetization from up to down occurs at the coercive field of the Co/Pd multilayer.
- This process is complicated but can be investigated using the EHE.
- Magnetization normal to the plane of the film is of increasing interest. However, it is difficult to reduce the noise of the magnetizations and keep it stable. The source of this noise is not understood and believed to come from multiple sources. This experiment looks into the magnetic after effect and its contribution to the magnetic noise that is seen in Co/Pd multilayer samples.

HYPOTHESIS

- The magnetic noise that was previously seen when switching the sample's magnetization is, in fact, not magnetic noise but is a remnant of the magnetic after effect due to its logarithmic time dependence.

METHOD

- An etched Co/Pd multilayer thin film (as shown below) allowed the EHE voltage to be measured.

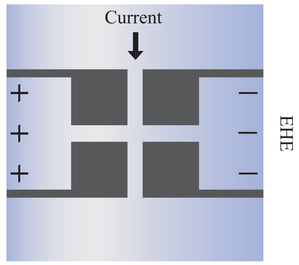


Figure 3. The above is a sample of a Co/Pd multilayer film. The sample was etched in a way that allowed the current to flow and the electrons to get deflected into seperate sections to allow the EHE to be measured.

- The sample was inserted between the faces of an electro-magnet. By sweeping the applied field a hysteresis loop was measured to determine the coercive fields (showed below).
- The applied field was cycled between points labled 1 - 4.
- The perpendicular magnetization of the sample was observed by these field sweeps and the time dependence of the magnetic after effect measured at several fields during the switching.

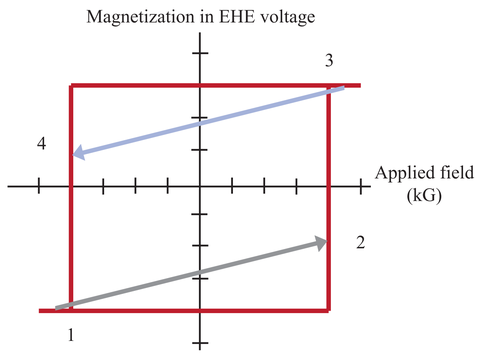


Figure 4. The above field depicts an ideal hysteresis loop of a Co/Pd. The applied field sweeps from a negative field to a positive field. Fields labeled as 2 and 4 are the coercive fields. These are the points where the magnetization changes at a specific point of the applied field. The experiment runs trials from points in saturation (1 and 4) to points in the coercive field.

RESULTS

- The drift through the cycle can be removed in the data analysis (described later).
- An average was taken for the cycles, and the decay rates at points 2 and 4 was analyzed.

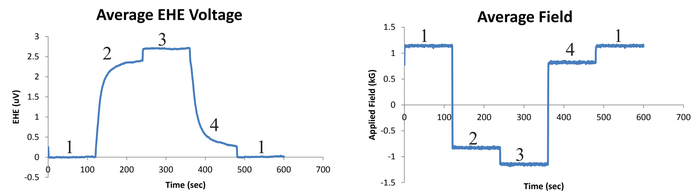


Figure 5. The image above depicts one cycle of the applied field. Points 2 and 4 are coercive points and the exponential time dependence is observed.

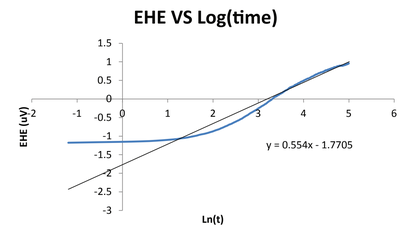


Figure 6. The above depicts the time relationship of the magnetic after effect on a log(time) plot. This relationship was determined by the averages of the cycles and reduction of the drift.

- Subtracting decay rate 4 from 2 and dividing by 2, eliminates the drift to give a time relationship described by a log(time) plot (an approximate log dependence is expected).

DISCUSSION

- These results show that the magnetic after effect behaves logarithmically in time.
- This helps in understanding the sources of magnetic noise.
- Understanding magnetic noise helps to further increase the stablility of hard drive bits and therefore memory storage.

ACKNOWLEDGEMENTS

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